

Performance Study

- Objective: Evaluate the texture and gloss performance of MicroTex 950 vs. Fluo 750TX in various powder coating systems

Physical Property:	MicroTex 950	Fluo 750TX
Wax Type	Modified polyolefin	PTFE
Melting Point °C	135 - 140	>325
Density @ 25 °C (g/cc)	1.22	2.20
Maximum particle size (µm)	74	145

Control Formulas

- Resins:
 - Epoxy-Polyester Hybrid
 - Polyester-TGIC
 - Polyester-HAA (Primid)
- Each system was evaluated with 1.0% Fluo 750TX vs. 4-5-6% MicroTex 950

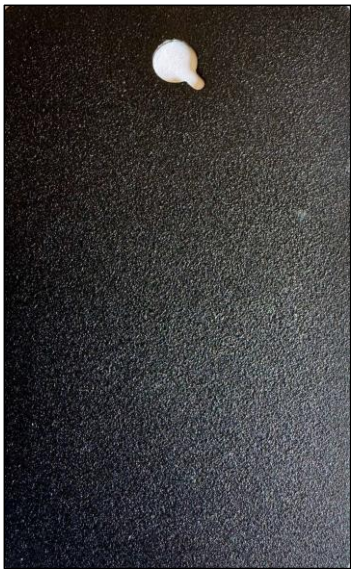
Material	Description	1% PTFE	4% MTX-950	5% MTX-950	6% MTX-950
Resin/crosslinker (See Results table)	Stoichiometrically calculated for each resin chemistry				
Resiflow PL-200	Flow Agent	5.00	5.00	5.00	5.00
Benzoin	Degassing Agent	2.50	2.50	2.50	2.50
Raven 2500	Carbon Black	5.00	5.00	5.00	5.00
Portaryte B10	Barium Sulfate	145.00	130.00	125.00	120.00
Fluo 750TX	PTFE	5.00			
MicroTex 950	Modified polyolefin		20.00	25.00	30.00

Test Methods Beyond the Visual

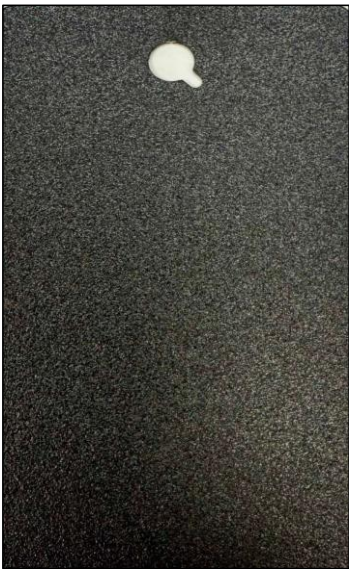
- Specular Gloss (ASTM D2457)
 - 60° geometry
- Impact Resistance (ASTM D2794) “falling weight test”
 - Measures resistance of a paint film from rapid deformation (inch/lb.)
- Conical Mandrel Bend (ASTM D522)
 - Measure of flexibility (crack resistance) 130°
- Crosshatch Adhesion (ASTM D3359)
 - Tape test to measure adhesion
- Pellet Flow (ASTM D3541)
 - Measurement melt flow under low shear

Visually our best replacement for Fluo 750TX

1.0% Fluo 750TX Hybrid



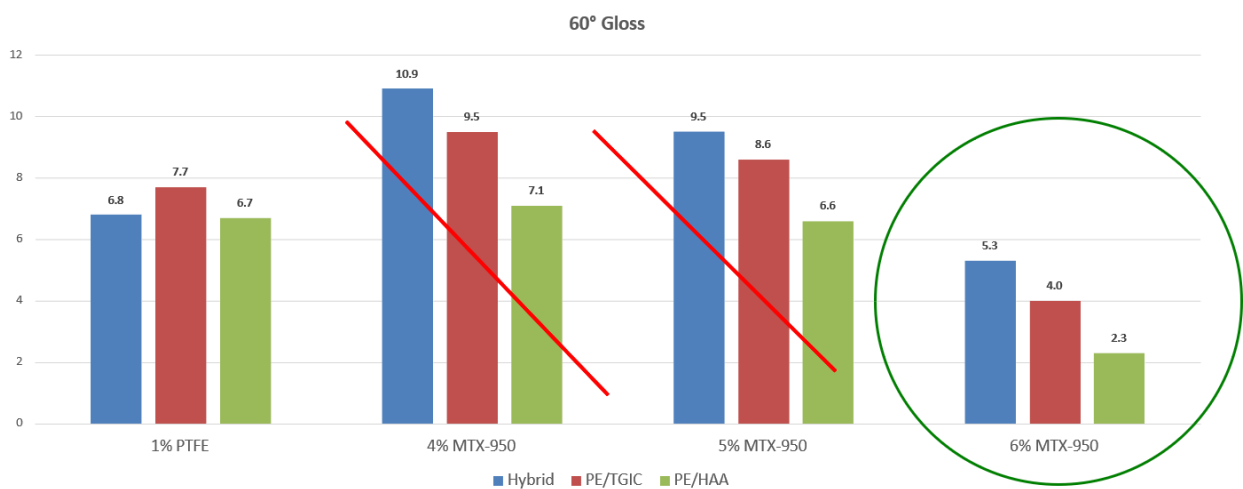
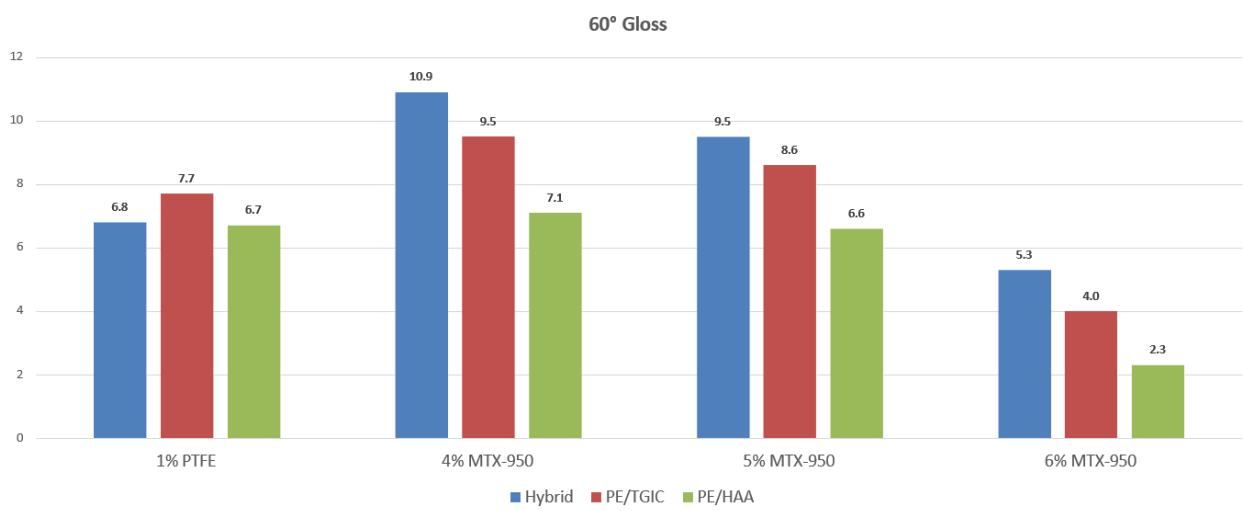
6.0% MicroTex 950 - Polyester TGIC



6.0% MicroTex 950 - Polyester HAA



Gloss Efficiency



Mechanical Properties

#	Chemistry	Description	Crosshatch Adhesion	Impact Resistance Inch/Pound		Conical Mandrel	Pellet Flow
				Forward	Reverse		
1	Epoxy Hybrid	1% Fluo 750TX	5B	100	160	Pass	16
2		4% MTX-950	5B	100	100	Pass	18
3		5% MTX-950	5B	160	100	Pass	17
4		6% MTX-950	5B	140	160	Pass	17
5	PE/TGIC	1% Fluo 750TX	5B	160	100	Pass	14
6		4% MTX-950	5B	160	160	Pass	16
7		5% MTX-950	5B	120	100	Pass	14
8		6% MTX-950	5B	80	60	Pass	13
9	PE/HAA	1% Fluo 750TX	5B	160	120	Pass	26
10		4% MTX-950	5B	120	60	Pass	24
11		5% MTX-950	5B	100	80	Pass	22
12		6% MTX-950	5B	80	60	Pass	20

Inclined Plate – Pellet Flow at 15 min/200°C
(indication of edge coverage)



Conclusion

- Visually MicroTex 950 is the closest match to Fluo 750TX
- Single digit gloss (<6 gloss units/60°)
- No negative effect on cross hatch adhesion or mandrel bend
- Impact resistance was slightly lower in the polyesters versus the epoxy hybrid
- Pellet flow indicates minor restricted flow; good for appearance and edge coverage in texture finishes
- We are receiving very positive customer feedback and approvals worldwide



France: **Axalta**



MicroTex 950
Modified polyolefin

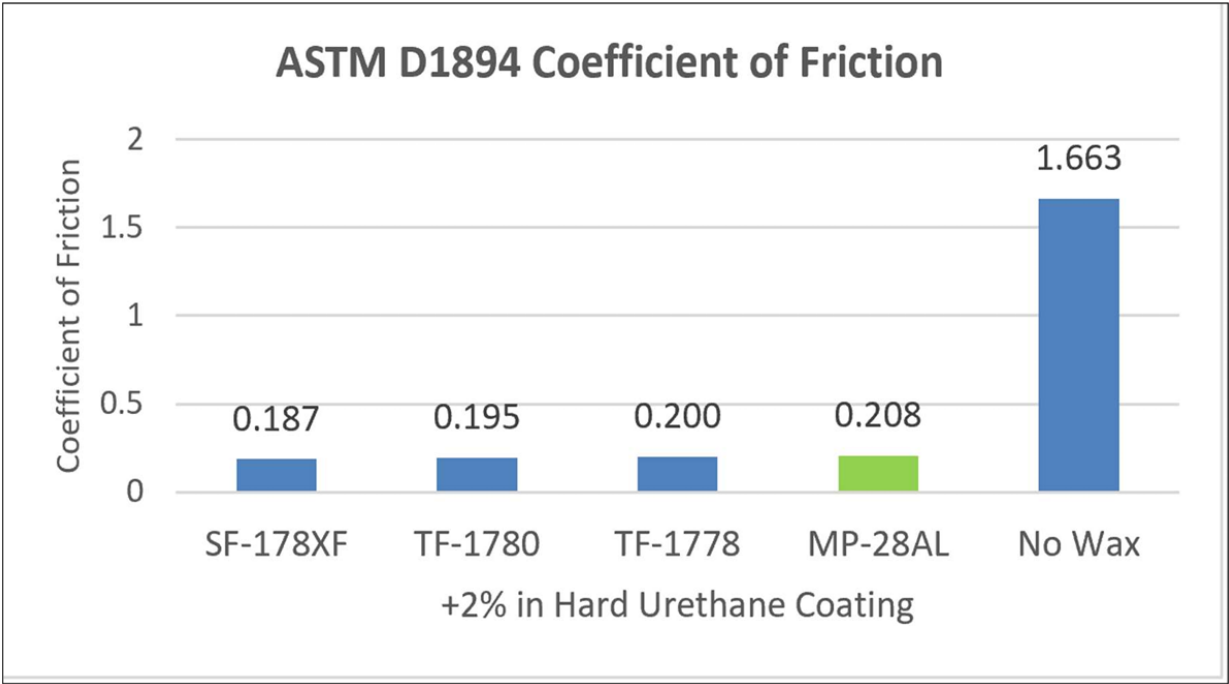
Powder coatings –
polyester / HAA

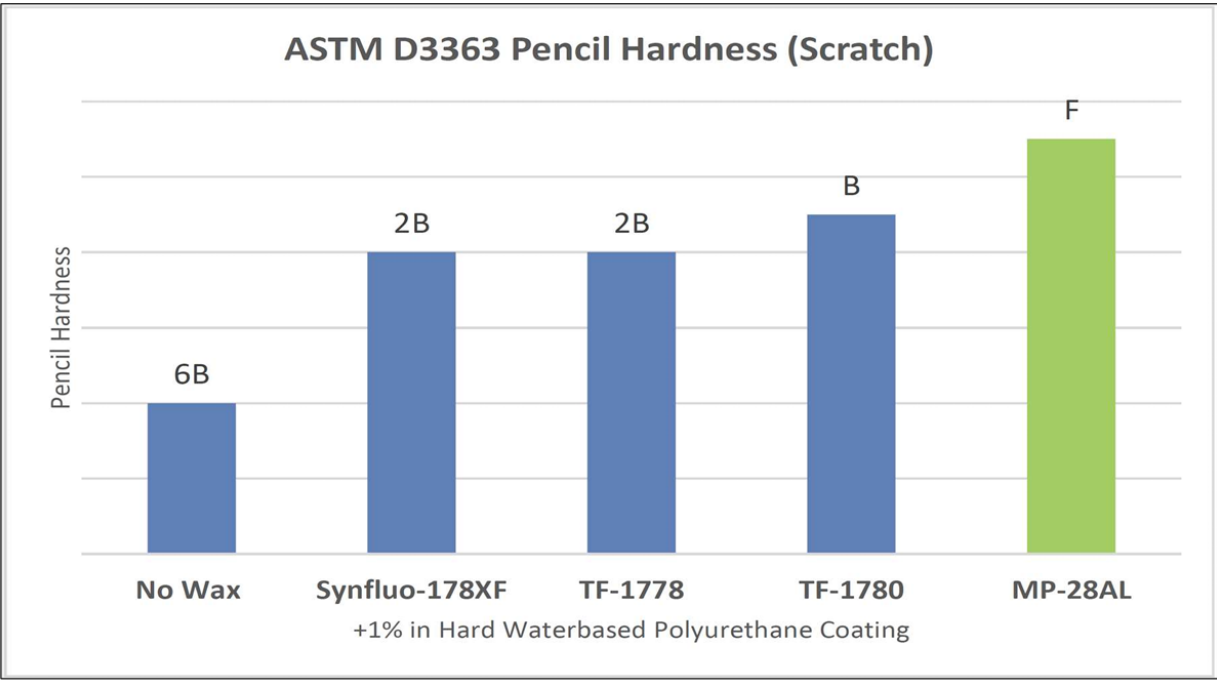
PTFE replacement /
consistent texture effect
with matting

MP-28AL vs. Polyolefin Wax-PTFE

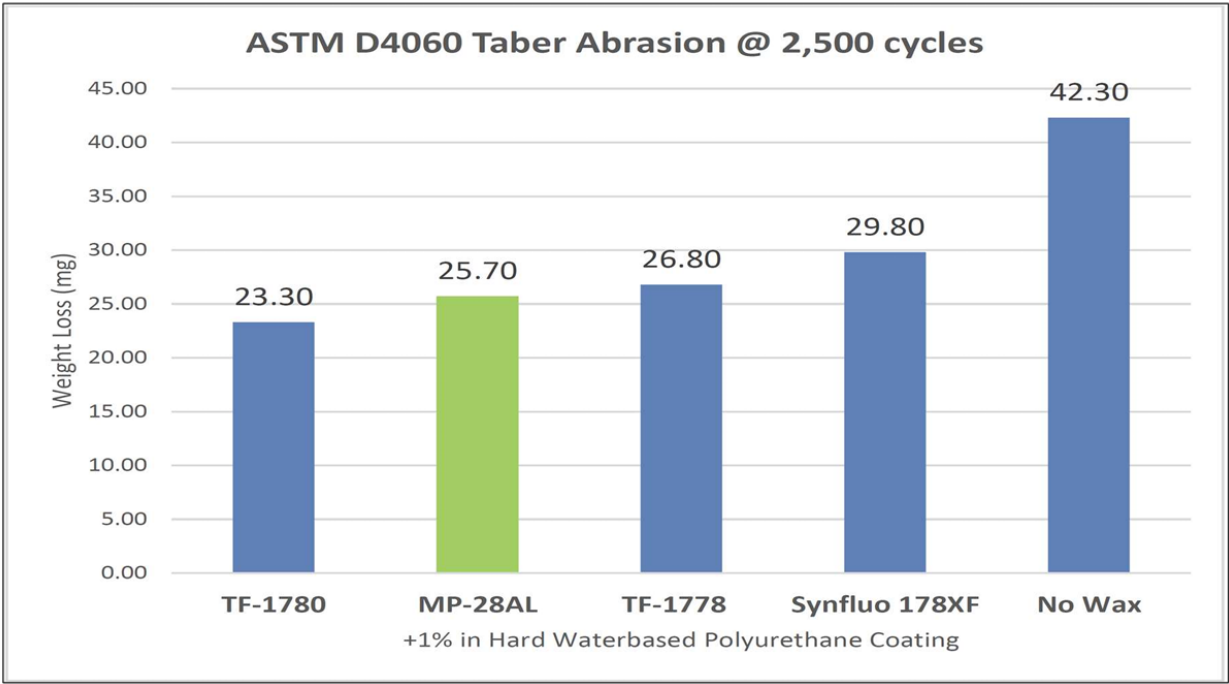
- **Objective:** Compare MP-28AL: Synfluo 178XF, Lanco TF -1778 and TF1780
 - 1.0% TFW - 1K aqueous polyurethane

Physical Property:	Micro Powders Synfluo 178XF	Lubrizol Lanco TF-1778	Lubrizol Lanco TF-1780
Wax Type	Synthetic wax/PTFE	PE/PTFE	PE/PTFE
Melting Point °C	104 – 110	102	102
Density at 25 °C (g/cc)	0.98	1.04	1.07
Mean particle size (µm)	3.0 – 5.0	~6.0	~5.0
Maximum particle size (µm)	15.56 (D100)	14.00 (D90)	10.00 (D90)





Pencil Scratch Hardness Rating													
Soft					Mid-Point				Hard				
8B	7B	6B	5B	4B	3B	2B	B	HB	F	H	2H	3H	4H



案例

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